SOMAIYA VIDYAVIHAR UNIVERSITY — K J SOMAIYA COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science | Academic Year 2026–2027

MEMBER CONTRIBUTION STATEMENT & WORK DIVISION REPORT

Course: Statistical Methods & Probability (IA-1) **Theme:** STAT-AI Engineering Challenge	**Case Study 1:** AI Heatwave Monitoring System **Evaluation:** 20 Marks (Live Showcase & Viva)	**GitHub:** github.com/varad-op/smp-maths-app **Live App:** Streamlit 6-Tab Multi-Module App

1. Individual Member Allocation & Contribution Summary

To ensure balanced participation and individual academic rigor, the 6-member team structured the project into modular, independent syllabus topics. Each student implemented their own statistical mathematical calculations, created interactive visualizations in their dedicated web application tab, and contributed their verified numerical steps to the investigation sheet.

Member & Slot	Designated Role	Assigned Syllabus Scope & Mathematical Focus	Primary Code & Artifacts	Share
Member 1 (Min 0:00–1:00)	Data Engineering & Ingestion Lead	• IMD & NASA POWER dataset ingestion (N = 65 observations). • Interquartile Range (IQR = 3.10°C) outlier screening. • 7-bin continuous frequency distribution table & cumulative Ogive.	`modules/data_loader.py` `app.py` (Tab 1)	**16.67%**
Member 2 (Rajashree KJ) (Min 1:00-2:00)	Descriptive Statistics Lead	• Central tendency: Mean (37.53 °C), Median (37.30 °C), Mode (36.80 °C). • Dispersion: Sample Variance (7.44 (°C)^2), Std Dev (2.73 °C), CV (7.27%). • Discretization check between ungrouped and grouped mean (Delta = 0.10 °C). • Box-and-Whisker temperature volatility analysis.	`modules/descriptive_stats.py` `app.py` (Tab 2)	**16.67%**
Member 3 (Min 2:00-3:15)	Probability & Extreme Values Lead	• Gaussian fitting: X ~ N(mu = 37.53 °C, sigma = 2.73 °C). • Standardized Z-scores for 38 °C, 40 °C, 42 °C, and 45 °C. • Analytical tail exceedance probabilities P(X >= T) via standard normal CDF. • Interactive threshold slider and bell curve tail shading.	`modules/extreme_value_analysis.py` `app.py` (Tab 3)	**16.67%**
Member 4 (Min 3:15-4:30)	Bayesian Inference Specialist	• 2x2 Joint and marginal contingency table (Temperature vs Humidity). • Climatological discovery: P(Heatwave | High Humidity) = 0.00% in Mumbai. • Bayes' Theorem IoT false-alarm filtration: updates P(H) = 10.77% to P(H|S) = 58.12% upon sensor trigger.	`modules/bayesian_inference.py` `app.py` (Tab 4)	**16.67%**
Member 5 (Samartha / venice067) (Min 4:30-5:30)	Time Series & Random Process Lead	• Discrete random process modeling {X_t} over 65 sequential days. • Sample Autocorrelation Function (ACF) for lags 1 through 7 (r1 = 0.9026). • Bartlett's 95% confidence bounds (±0.2431) for significance testing. • Detection of 6-day heatwave streak (May 4-9) and 7-day rolling stationarity.	`modules/time_series_process.py` `app.py` (Tab 5)	**16.67%**
Member 6 (Varad Bele - Lead) (Min 5:30-6:45)	AI Decision Engine & UI Lead	• Multi-Tier IMD Decision Matrix (Green, Yellow, Orange, Red alerts). • Live simulation sliders, microclimate zones, and 1-click historical presets. • Municipal impact estimator (hospital triage beds, water tankers, power grid). • Master GitHub repository management, code integration, and UI design.	`modules/decision_engine.py` `app.py` (Tab 6, Lead UI)	**16.67%**
Total Project	**Consolidated Team Effort**	**Comprehensive statistical investigation and live interactive web system**	**6 Integrated Modules**	**100.0%**

2. Collaborative Git Workflow Verification

• **Version Control:** Maintained on GitHub under repository `https://github.com/varad-op/smp-maths-app`.
• **Branching Strategy:** Each member was allocated an isolated branch (`member-X-...`) to prevent merge conflicts and maintain atomic commit logs representing their individual code contributions.
• **Live Application Architecture:** Modular multi-tab Streamlit dashboard allowing continuous live demonstration of all six members' computational pipelines.

3. Team Declaration & Sign-Off

We hereby declare that this Internal Assessment (IA-1) submission represents our genuine, collaborative engineering work. All calculations were computed from verified observational meteorological data, and every team member contributed equally to their designated theoretical and software components.

Varad Bele	**Rajashree KJ**	**Samartha**	**Member 1, 3, 4**
Team Lead (Member 6)	Member 2	Member 5	Team Members